

Q. 1 – Q. 5 carry one mark each.

Q.1 The man who is now Municipal Commissioner worked as _____.

- (A) the security guard at a university
- (B) a security guard at the university
- (C) a security guard at university
- (D) the security guard at the university

Q.2 Nobody knows how the Indian cricket team is going to cope with the difficult and seamer-friendly wickets in Australia.

Choose the option which is closest in meaning to the underlined phrase in the above sentence.

- (A) put up with (B) put in with (C) put down to (D) put up against

Q.3 Find the odd one in the following group of words.

mock, deride, praise, jeer

- (A) mock (B) deride (C) praise (D) jeer

Q.4 Pick the odd one from the following options.

- (A) CADBE (B) JHKIL (C) XWYZ (D) ONPMQ

Q.5 In a quadratic function, the value of the product of the roots (α, β) is 4. Find the value of

$$\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}}$$

- (A) n^4 (B) 4^n (C) 2^{2n-1} (D) 4^{n-1}

Q. 6 – Q. 10 carry two marks each.

Q.6 Among 150 faculty members in an institute, 55 are connected with each other through Facebook® and 85 are connected through WhatsApp®. 30 faculty members do not have Facebook® or WhatsApp® accounts. The number of faculty members connected only through Facebook® accounts is _____.

- (A) 35 (B) 45 (C) 65 (D) 90

- Q.7 Computers were invented for performing only high-end useful computations. However, it is no understatement that they have taken over our world today. The internet, for example, is ubiquitous. Many believe that the internet itself is an unintended consequence of the original invention. With the advent of mobile computing on our phones, a whole new dimension is now enabled. One is left wondering if all these developments are good or, more importantly, required.

Which of the statement(s) below is/are logically valid and can be inferred from the above paragraph?

- (i) The author believes that computers are not good for us.
- (ii) Mobile computers and the internet are both intended inventions

(A) (i) only (B) (ii) only (C) both (i) and (ii) (D) neither (i) nor (ii)

- Q.8 All hill-stations have a lake. Ooty has two lakes.

Which of the statement(s) below is/are logically valid and can be inferred from the above sentences?

- (i) Ooty is not a hill-station.
- (ii) No hill-station can have more than one lake.

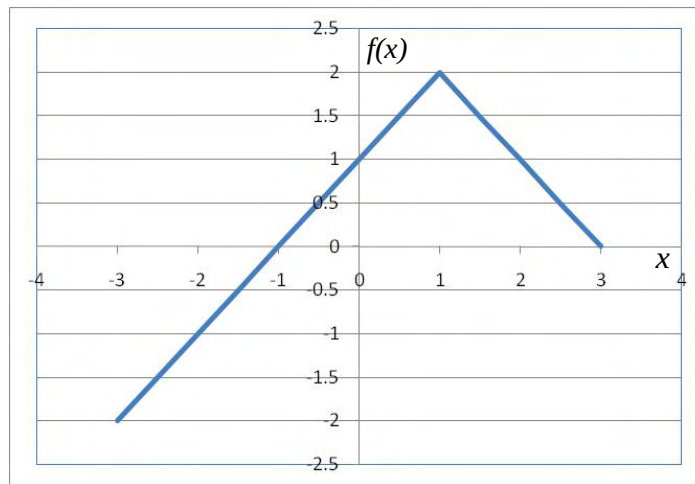
(A) (i) only (B) (ii) only
(C) both (i) and (ii) (D) neither (i) nor (ii)

- Q.9 In a 2×4 rectangle grid shown below, each cell is a rectangle. How many rectangles can be observed in the grid?



(A) 21 (B) 27 (C) 30 (D) 36

Q.10



Choose the correct expression for $f(x)$ given in the graph.

(A) $f(x) = 1 - |x - 1|$

(B) $f(x) = 1 + |x - 1|$

(C) $f(x) = 2 - |x - 1|$

(D) $f(x) = 2 + |x - 1|$

END OF THE QUESTION PAPER

Q. 1 - Q. 25 carry one mark each.

Q.1 Consider the following expressions:

- (i) *false*
- (ii) Q
- (iii) *true*
- (iv) $P \vee Q$
- (v) $\neg Q \vee P$

The number of expressions given above that are logically implied by $P \wedge (P \Rightarrow Q)$ is _____ .

Q.2 Let $f(x)$ be a polynomial and $g(x) = f'(x)$ be its derivative. If the degree of $(f(x) + f(-x))$ is 10, then the degree of $(g(x) - g(-x))$ is _____ .

Q.3 The minimum number of colours that is sufficient to vertex-colour any planar graph is _____ .

Q.4 Consider the systems, each consisting of m linear equations in n variables.

- I. If $m < n$, then all such systems have a solution
- II. If $m > n$, then none of these systems has a solution
- III. If $m = n$, then there exists a system which has a solution

Which one of the following is **CORRECT**?

- (A) I, II and III are true
- (B) Only II and III are true
- (C) Only III is true
- (D) None of them is true

- Q.5** Suppose that a shop has an equal number of LED bulbs of two different types. The probability of an LED bulb lasting more than 100 hours given that it is of Type 1 is 0.7, and given that it is of Type 2 is 0.4. The probability that an LED bulb chosen uniformly at random lasts more than 100 hours is _____ .
- Q.6** Suppose that the eigenvalues of matrix A are 1, 2, 4. The determinant of $(A^{-1})^T$ is _____ .
- Q.7** Consider an eight-bit ripple-carry adder for computing the sum of A and B , where A and B are integers represented in 2's complement form. If the decimal value of A is one, the decimal value of B that leads to the longest latency for the sum to stabilize is _____ .
- Q.8** Let, $x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$ where x_1, x_2, x_3, x_4 are Boolean variables, and $\oplus$ is the XOR operator. Which one of the following must always be **TRUE**?
- (A) $x_1 x_2 x_3 x_4 = 0$
 - (B) $x_1 x_3 + x_2 = 0$
 - (C) $\bar{x}_1 \oplus \bar{x}_3 = \bar{x}_2 \oplus \bar{x}_4$
 - (D) $x_1 + x_2 + x_3 + x_4 = 0$
- Q.9** Let X be the number of distinct 16-bit integers in 2's complement representation. Let Y be the number of distinct 16-bit integers in sign magnitude representation. Then $X - Y$ is _____ .
- Q.10** A processor has 40 distinct instructions and 24 general purpose registers. A 32-bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is _____ .

Q.11 Breadth First Search (BFS) is started on a binary tree beginning from the root vertex. There is a vertex t at a distance four from the root. If t is the n -th vertex in this BFS traversal, then the maximum possible value of n is _____ .

Q.12 The value printed by the following program is _____ .

```
void f(int* p, int m){
    m = m + 5;
    *p = *p + m;
    return;
}

void main(){
    int i=5, j=10;

    f(&i, j);
    printf("%d", i+j);
}
```

Q.13 Assume that the algorithms considered here sort the input sequences in ascending order. If the input is already in ascending order, which of the following are **TRUE**?

- I. Quicksort runs in $\Theta(n^2)$ time
- II. Bubblesort runs in $\Theta(n^2)$ time
- III. Mergesort runs in $\Theta(n)$ time
- IV. Insertion sort runs in $\Theta(n)$ time

- (A) I and II only
- (B) I and III only
- (C) II and IV only
- (D) I and IV only

- Q.14** The Floyd-Warshall algorithm for all-pair shortest paths computation is based on
- (A) Greedy paradigm.
 - (B) Divide-and-Conquer paradigm.
 - (C) Dynamic Programming paradigm.
 - (D) neither Greedy nor Divide-and-Conquer nor Dynamic Programming paradigm.
- Q.15** N items are stored in a sorted doubly linked list. For a *delete* operation, a pointer is provided to the record to be deleted. For a *decrease-key* operation, a pointer is provided to the record on which the operation is to be performed.
- An algorithm performs the following operations on the list in this order: $\Theta(N)$ *delete*, $O(\log N)$ *insert*, $O(\log N)$ *find*, and $\Theta(N)$ *decrease-key*. What is the time complexity of all these operations put together?
- (A) $O(\log^2 N)$ (B) $O(N)$ (C) $O(N^2)$ (D) $\Theta(N^2 \log N)$
- Q.16** The number of states in the minimum sized DFA that accepts the language defined by the regular expression
- $$(0 + 1)^*(0 + 1)(0 + 1)^*$$
- is _____ .
- Q.17** Language L_1 is defined by the grammar: $S_1 \rightarrow aS_1b | \epsilon$
Language L_2 is defined by the grammar: $S_2 \rightarrow abS_2 | \epsilon$
- Consider the following statements:
- P : L_1 is regular
 Q : L_2 is regular
- Which one of the following is **TRUE**?
- (A) Both P and Q are true
 - (B) P is true and Q is false
 - (C) P is false and Q is true
 - (D) Both P and Q are false

Q.18 Consider the following types of languages: L_1 : Regular, L_2 : Context-free, L_3 : Recursive, L_4 : Recursively enumerable. Which of the following is/are **TRUE**?

- I. $\overline{L_3} \cup L_4$ is recursively enumerable
- II. $\overline{L_2} \cup L_3$ is recursive
- III. $L_1^* \cap L_2$ is context-free
- IV. $L_1 \cup \overline{L_2}$ is context-free

- (A) I only
- (B) I and III only
- (C) I and IV only
- (D) I, II and III only

Q.19 Match the following:

- | | |
|--------------------------|---------------------------|
| (P) Lexical analysis | (i) Leftmost derivation |
| (Q) Top down parsing | (ii) Type checking |
| (R) Semantic analysis | (iii) Regular expressions |
| (S) Runtime environments | (iv) Activation records |

- (A) $P \leftrightarrow i, Q \leftrightarrow ii, R \leftrightarrow iv, S \leftrightarrow iii$
- (B) $P \leftrightarrow iii, Q \leftrightarrow i, R \leftrightarrow ii, S \leftrightarrow iv$
- (C) $P \leftrightarrow ii, Q \leftrightarrow iii, R \leftrightarrow i, S \leftrightarrow iv$
- (D) $P \leftrightarrow iv, Q \leftrightarrow i, R \leftrightarrow ii, S \leftrightarrow iii$

Q.20 In which one of the following page replacement algorithms it is possible for the page fault rate to increase even when the number of allocated frames increases?

- (A) LRU (Least Recently Used)
- (B) OPT (Optimal Page Replacement)
- (C) MRU (Most Recently Used)
- (D) FIFO (First In First Out)

- Q.21** B+ Trees are considered **BALANCED** because
- (A) the lengths of the paths from the root to all leaf nodes are all equal.
 - (B) the lengths of the paths from the root to all leaf nodes differ from each other by at most 1.
 - (C) the number of children of any two non-leaf sibling nodes differ by at most 1.
 - (D) the number of records in any two leaf nodes differ by at most 1.
- Q.22** Suppose a database schedule S involves transactions $T_1, \dots, T_n$. Construct the precedence graph of S with vertices representing the transactions and edges representing the conflicts. If S is serializable, which one of the following orderings of the vertices of the precedence graph is guaranteed to yield a serial schedule?
- (A) Topological order
 - (B) Depth-first order
 - (C) Breadth-first order
 - (D) Ascending order of transaction indices
- Q.23** Anarkali digitally signs a message and sends it to Salim. Verification of the signature by Salim requires
- (A) Anarkali's public key.
 - (B) Salim's public key.
 - (C) Salim's private key.
 - (D) Anarkali's private key.
- Q.24** In an Ethernet local area network, which one of the following statements is **TRUE**?
- (A) A station stops to sense the channel once it starts transmitting a frame.
 - (B) The purpose of the jamming signal is to pad the frames that are smaller than the minimum frame size.
 - (C) A station continues to transmit the packet even after the collision is detected.
 - (D) The exponential backoff mechanism reduces the probability of collision on retransmissions.

Q.25 Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

- (A) HTTP GET request, DNS query, TCP SYN
- (B) DNS query, HTTP GET request, TCP SYN
- (C) DNS query, TCP SYN, HTTP GET request
- (D) TCP SYN, DNS query, HTTP GET request

Q. 26 - Q. 55 carry two marks each.

Q.26 A binary relation R on $\mathbb{N} \times \mathbb{N}$ is defined as follows: $(a, b)R(c, d)$ if $a \leq c$ or $b \leq d$. Consider the following propositions:

P: R is reflexive

Q: R is transitive

Which one of the following statements is **TRUE**?

- (A) Both P and Q are true.
- (B) P is true and Q is false.
- (C) P is false and Q is true.
- (D) Both P and Q are false.

Q.27 Which one of the following well-formed formulae in predicate calculus is **NOT** valid?

- (A) $(\forall x p(x) \Rightarrow \forall x q(x)) \Rightarrow (\exists x \neg p(x) \vee \forall x q(x))$
- (B) $(\exists x p(x) \vee \exists x q(x)) \Rightarrow \exists x (p(x) \vee q(x))$
- (C) $\exists x (p(x) \wedge q(x)) \Rightarrow (\exists x p(x) \wedge \exists x q(x))$
- (D) $\forall x (p(x) \vee q(x)) \Rightarrow (\forall x p(x) \vee \forall x q(x))$

Q.28 Consider a set U of 23 different compounds in a Chemistry lab. There is a subset S of U of 9 compounds, each of which reacts with exactly 3 compounds of U . Consider the following statements:

- I. Each compound in $U \setminus S$ reacts with an odd number of compounds.
- II. At least one compound in $U \setminus S$ reacts with an odd number of compounds.
- III. Each compound in $U \setminus S$ reacts with an even number of compounds.

Which one of the above statements is **ALWAYS TRUE**?

- (A) Only I
- (B) Only II
- (C) Only III
- (D) None

Q.29 The value of the expression $13^{99} \pmod{17}$, in the range 0 to 16, is _____.

Q.30 Suppose the functions F and G can be computed in 5 and 3 nanoseconds by functional units U_F and U_G , respectively. Given two instances of U_F and two instances of U_G , it is required to implement the computation $F(G(X_i))$ for $1 \leq i \leq 10$. Ignoring all other delays, the minimum time required to complete this computation is _____ nanoseconds.

Q.31 Consider a processor with 64 registers and an instruction set of size twelve. Each instruction has five distinct fields, namely, opcode, two source register identifiers, one destination register identifier, and a twelve-bit immediate value. Each instruction must be stored in memory in a byte-aligned fashion. If a program has 100 instructions, the amount of memory (in bytes) consumed by the program text is _____.

Q.32 The width of the physical address on a machine is 40 bits. The width of the tag field in a 512 KB 8-way set associative cache is _____ bits.

- Q.33** Consider a 3 GHz (gigahertz) processor with a three-stage pipeline and stage latencies τ_1 , τ_2 , and τ_3 such that $\tau_1 = 3\tau_2/4 = 2\tau_3$. If the longest pipeline stage is split into two pipeline stages of equal latency, the new frequency is _____ GHz, ignoring delays in the pipeline registers.
- Q.34** A complete binary min-heap is made by including each integer in $[1, 1023]$ exactly once. The depth of a node in the heap is the length of the path from the root of the heap to that node. Thus, the root is at depth 0. The maximum depth at which integer 9 can appear is _____.
- Q.35** The following function computes X^Y for positive integers X and Y.

```
int exp(int X, int Y) {
    int res = 1, a = X, b = Y;

    while ( b != 0 ) {
        if ( b%2 == 0 ) { a = a*a; b = b/2; }
        else          { res = res*a; b = b-1; }
    }
    return res;
}
```

Which one of the following conditions is **TRUE** before every iteration of the loop?

- (A) $X^Y = a^b$
- (B) $(res * a)^Y = (res * X)^b$
- (C) $X^Y = res * a^b$
- (D) $X^Y = (res * a)^b$

Q.36 Consider the following New-order strategy for traversing a binary tree:

- Visit the root;
- Visit the right subtree using New-order;
- Visit the left subtree using New-order;

The New-order traversal of the expression tree corresponding to the reverse polish expression $3\ 4\ *\ 5\ -\ 2\ ^\ 6\ 7\ *\ 1\ +\ -$ is given by:

- (A) $+ - 1\ 6\ 7\ *\ 2\ ^\ 5 - 3\ 4\ *$
(B) $- + 1\ *\ 6\ 7\ ^\ 2 - 5\ *\ 3\ 4$
(C) $- + 1\ *\ 7\ 6\ ^\ 2 - 5\ *\ 4\ 3$
(D) $1\ 7\ 6\ *\ +\ 2\ 5\ 4\ 3\ *\ -\ ^\ -$

Q.37 Consider the following program:

```
int f(int *p, int n)
{
    if (n <= 1) return 0;
    else return max(f(p+1,n-1),p[0]-p[1]);
}

int main()
{
    int a[] = {3,5,2,6,4};
    printf("%d", f(a,5));
}
```

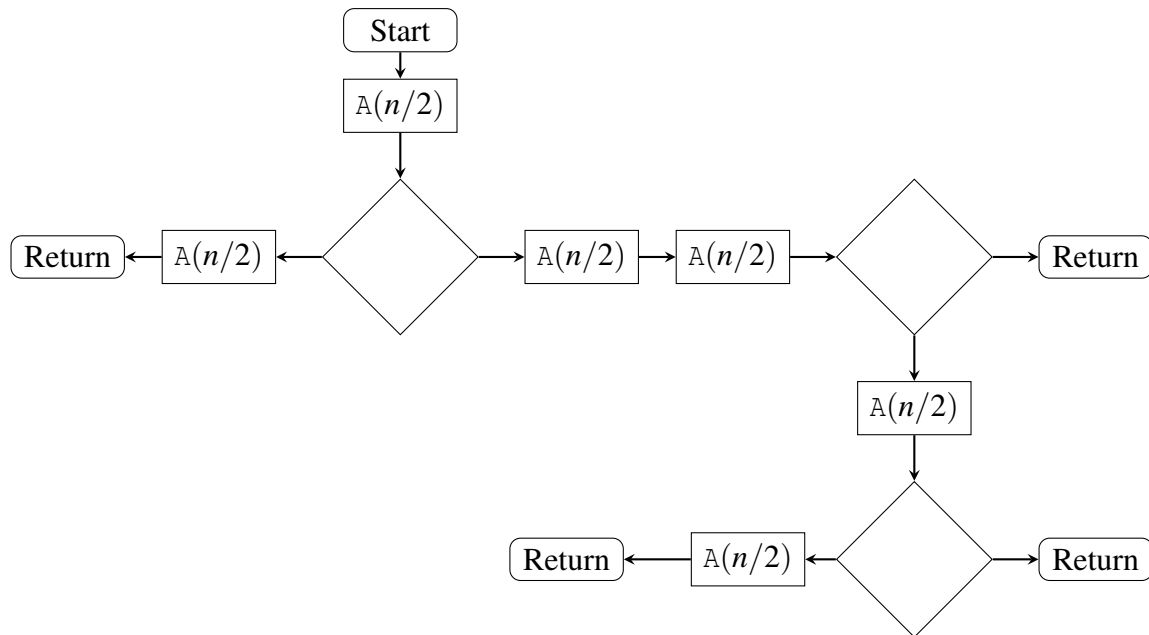
Note: $\max(x, y)$ returns the maximum of x and y .

The value printed by this program is _____ .

Q.38 Let A_1, A_2, A_3 , and A_4 be four matrices of dimensions $10 \times 5, 5 \times 20, 20 \times 10$, and 10×5 , respectively. The minimum number of scalar multiplications required to find the product $A_1 A_2 A_3 A_4$ using the basic matrix multiplication method is _____ .

- Q.39** The given diagram shows the flowchart for a recursive function $A(n)$. Assume that all statements, except for the recursive calls, have $O(1)$ time complexity. If the worst case time complexity of this function is $O(n^\alpha)$, then the least possible value (accurate up to two decimal positions) of α is _____.

Flowchart for Recursive Function $A(n)$



- Q.40** The number of ways in which the numbers 1, 2, 3, 4, 5, 6, 7 can be inserted in an empty binary search tree, such that the resulting tree has height 6, is _____ .
- Note: The height of a tree with a single node is 0.*

Q.41 In an adjacency list representation of an undirected simple graph $G = (V, E)$, each edge (u, v) has two adjacency list entries: $[v]$ in the adjacency list of u , and $[u]$ in the adjacency list of v . These are called twins of each other. A twin pointer is a pointer from an adjacency list entry to its twin. If $|E| = m$ and $|V| = n$, and the memory size is not a constraint, what is the time complexity of the most efficient algorithm to set the twin pointer in each entry in each adjacency list?

- (A) $\Theta(n^2)$
- (B) $\Theta(n + m)$
- (C) $\Theta(m^2)$
- (D) $\Theta(n^4)$

Q.42 Consider the following two statements:

- I.** If all states of an NFA are accepting states then the language accepted by the NFA is Σ^* .
- II.** There exists a regular language A such that for all languages B , $A \cap B$ is regular.

Which one of the following is **CORRECT**?

- (A) Only **I** is true
- (B) Only **II** is true
- (C) Both **I** and **II** are true
- (D) Both **I** and **II** are false

Q.43 Consider the following languages:

$$L_1 = \{a^n b^m c^{n+m} : m, n \geq 1\}$$

$$L_2 = \{a^n b^n c^{2n} : n \geq 1\}$$

Which one of the following is **TRUE**?

- (A) Both L_1 and L_2 are context-free.
- (B) L_1 is context-free while L_2 is not context-free.
- (C) L_2 is context-free while L_1 is not context-free.
- (D) Neither L_1 nor L_2 is context-free.

Q.44 Consider the following languages.

$$\begin{aligned} L_1 &= \{\langle M \rangle \mid M \text{ takes at least 2016 steps on some input}\}, \\ L_2 &= \{\langle M \rangle \mid M \text{ takes at least 2016 steps on all inputs}\} \text{ and} \\ L_3 &= \{\langle M \rangle \mid M \text{ accepts } \varepsilon\}, \end{aligned}$$

where for each Turing machine M , $\langle M \rangle$ denotes a specific encoding of M . Which one of the following is **TRUE**?

- (A) L_1 is recursive and L_2, L_3 are not recursive
- (B) L_2 is recursive and L_1, L_3 are not recursive
- (C) L_1, L_2 are recursive and L_3 is not recursive
- (D) L_1, L_2, L_3 are recursive

Q.45 Which one of the following grammars is free from *left recursion*?

- (A)
$$\begin{aligned} S &\rightarrow AB \\ A &\rightarrow Aa \mid b \\ B &\rightarrow c \end{aligned}$$
- (B)
$$\begin{aligned} S &\rightarrow Ab \mid Bb \mid c \\ A &\rightarrow Bd \mid \varepsilon \\ B &\rightarrow e \end{aligned}$$
- (C)
$$\begin{aligned} S &\rightarrow Aa \mid B \\ A &\rightarrow Bb \mid Sc \mid \varepsilon \\ B &\rightarrow d \end{aligned}$$
- (D)
$$\begin{aligned} S &\rightarrow Aa \mid Bb \mid c \\ A &\rightarrow Bd \mid \varepsilon \\ B &\rightarrow Ae \mid \varepsilon \end{aligned}$$

- Q.46** A student wrote two context-free grammars **G1** and **G2** for generating a single C-like array declaration. The dimension of the array is at least one. For example,

```
int a[10][3];
```

The grammars use D as the start symbol, and use six terminal symbols **int ; id [] num**.

Grammar **G1**

$D \rightarrow \text{int}L;$

$L \rightarrow \text{id}[E$

$E \rightarrow \text{num}]$

$E \rightarrow \text{num}][E$

Grammar **G2**

$D \rightarrow \text{int}L;$

$L \rightarrow \text{id}E$

$E \rightarrow E[\text{num}]$

$E \rightarrow [\text{num}]$

Which of the grammars correctly generate the declaration mentioned above?

- (A) Both **G1** and **G2**
- (B) Only **G1**
- (C) Only **G2**
- (D) Neither **G1** nor **G2**

- Q.47** Consider the following processes, with the arrival time and the length of the CPU burst given in milliseconds. The scheduling algorithm used is preemptive shortest remaining-time first.

Process	Arrival Time	Burst Time
P_1	0	10
P_2	3	6
P_3	7	1
P_4	8	3

The average turn around time of these processes is _____ milliseconds.

Q.48 Consider the following two-process synchronization solution.

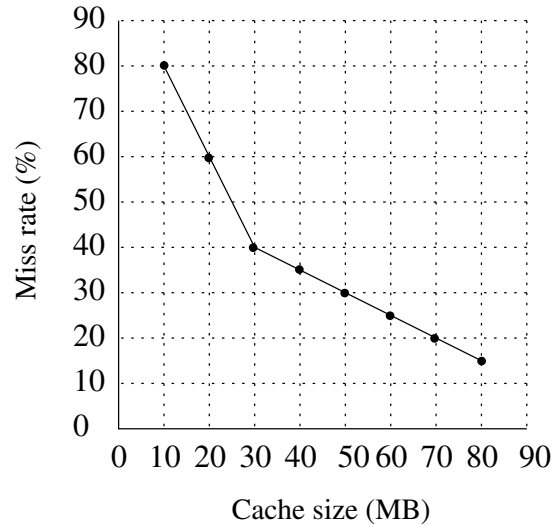
Process 0	Process 1
-----	-----
Entry: loop while (turn == 1);	Entry: loop while (turn == 0);
(critical section)	(critical section)
Exit: turn = 1;	Exit: turn = 0;

The shared variable `turn` is initialized to zero. Which one of the following is **TRUE**?

- (A) This is a correct two-process synchronization solution.
- (B) This solution violates mutual exclusion requirement.
- (C) This solution violates progress requirement.
- (D) This solution violates bounded wait requirement.

Q.49 Consider a non-negative counting semaphore S . The operation $P(S)$ decrements S , and $V(S)$ increments S . During an execution, 20 $P(S)$ operations and 12 $V(S)$ operations are issued in some order. The largest initial value of S for which at least one $P(S)$ operation will remain blocked is _____.

- Q.50** A file system uses an in-memory cache to cache disk blocks. The miss rate of the cache is shown in the figure. The latency to read a block from the cache is 1 ms and to read a block from the disk is 10 ms. Assume that the cost of checking whether a block exists in the cache is negligible. Available cache sizes are in multiples of 10 MB.



The smallest cache size required to ensure an average read latency of less than 6 ms is _____ MB.

- Q.51** Consider the following database schedule with two transactions, T_1 and T_2 .

$$S = r_2(X); r_1(X); r_2(Y); w_1(X); r_1(Y); w_2(X); a_1; a_2$$

where $r_i(Z)$ denotes a *read* operation by transaction T_i on a variable Z , $w_i(Z)$ denotes a *write* operation by T_i on a variable Z and a_i denotes an *abort* by transaction T_i .

Which one of the following statements about the above schedule is **TRUE**?

- (A) S is non-recoverable
- (B) S is recoverable, but has a cascading abort
- (C) S does not have a cascading abort
- (D) S is strict

Q.52 Consider the following database table named *water_schemes* :

<i>water_schemes</i>		
scheme_no	district_name	capacity
1	Ajmer	20
1	Bikaner	10
2	Bikaner	10
3	Bikaner	20
1	Churu	10
2	Churu	20
1	Dungargarh	10

The number of tuples returned by the following SQL query is _____ .

```
with total(name, capacity) as
  select district_name, sum(capacity)
  from water_schemes
  group by district_name
with total_avg(capacity) as
  select avg(capacity)
  from total
select name
  from total, total_avg
  where total.capacity ≥ total_avg.capacity
```

Q.53 A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is _____ bytes.

Q.54 For the IEEE 802.11 MAC protocol for wireless communication, which of the following statements is/are **TRUE**?

- I. At least three non-overlapping channels are available for transmissions.
- II. The RTS-CTS mechanism is used for collision detection.
- III. Unicast frames are ACKed.

- (A) All I, II, and III
- (B) I and III only
- (C) II and III only
- (D) II only

Q.55 Consider a 128×10^3 bits/second satellite communication link with one way propagation delay of 150 milliseconds. Selective retransmission (repeat) protocol is used on this link to send data with a frame size of 1 kilobyte. Neglect the transmission time of acknowledgement. The minimum number of bits required for the sequence number field to achieve 100% utilization is _____ .